

Exploring the Association Between Education and Health Quality: A Country Level Study

Final Report

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1. Visualization Matrix and Conclusions

Research question: Is there an association between education and population health quality?

Serial #	Hypothesis	Question	Analytic type	Variable(s)	Chart	Conclusion
1	Higher educational attainment is associated with lower mortality across countries.	How does Bachelor's attainment correlate with crude death rate across countries?	Predictive	Country; Educational attainment, at least Bachelor's or equivalent; Death rate, crude	Scatter plot with trend	Bachelor's attainment showed a weak positive association with crude death rate, likely driven by aging populations in high-attainment European countries.
2	Higher educational attainment is associated with lower mortality across countries.	How has crude death rate trended across youth NEET bands from 2010-2019?	Descriptive	Country; Year; Share of youth NEET; Death rate, crude	Multi-line chart	Crude death rate trends showed no consistent differentiation across youth NEET bands from 2012–2019, suggesting mortality was driven by factors beyond educational engagement alone.
3	Higher educational attainment is associated with lower mortality across countries.	Does crude death rate distribution differ by compulsory education duration bands?	Descriptive	Country; Year; Compulsory education, duration; Death rate, crude	Multi-box plot	Crude death rate distributions showed minimal variation across compulsory education duration bands, indicating a limited association with

						mortality outcomes.
4	Higher educational attainment is associated with lower mortality across countries.	How does tertiary school life expectancy relate to death rate by population?	Predictive; Descriptive	Country; Population; School life expectancy, tertiary; Death rate, crude	Bubble chart	Tertiary school life expectancy showed a weak positive association with crude death rate, with regional and population differences accounting for much of the variation observed.
5	Education levels shape reproductive and child health outcomes.	How does Bachelor's attainment relate to fertility rate at the country level across GDP per capita bands?	Predictive	Country; Educational attainment, at least Bachelor's or equivalent; Fertility rate; GDP per capita	Multi-scatter plot with trends	Bachelor's attainment showed a strong negative association with fertility rate in low GDP countries, with the relationship weakening significantly at higher income levels.
6	Education levels shape reproductive and child health outcomes.	How does DPT immunization coverage compare across tertiary school life expectancy bands?	Descriptive	Country; School life expectancy, tertiary; Immunization DPT	Bar chart	DPT immunization coverage increased consistently with tertiary school life expectancy, supporting the association between higher education access and improved child health outcomes.

7	Education levels shape reproductive and child health outcomes.	How has fertility rate changed over time by Bachelor's attainment bands?	Descriptive; Predictive (can see trend)	Country; Year; Educational attainment, at least Bachelor's or equivalent; Fertility rate, total	Area chart	Fertility rates were consistently highest in low attainment countries and lowest in high attainment countries, supporting the association between higher education and reduced fertility rates over time.
8	Education levels shape reproductive and child health outcomes.	How does compulsory education duration relate to fertility rates across counties?	Predictive	Country; Compulsory education, duration; Fertility rate, total; Region	Scatter plot with trend	Longer compulsory education duration was associated with lower fertility rates, with Sub-Saharan Africa showing the highest fertility rates regardless of duration.
9	Education reduces disease burden.	How does TB incidence vary across Bachelor's attainment bands by region?	Descriptive	Country; Region; Educational attainment, at least Bachelor's or equivalent; Incidence of tuberculosis	Tree map	TB incidence was highest in regions with low Bachelor's attainment, particularly Sub-Saharan Africa and South Asia, while higher attainment bands were associated with smaller block sizes, supporting the association between educational attainment and

						reduced disease burden.
10	Education reduces disease burden.	How does TB incidence vary across youth NEET rate bands by select regions?	Descriptive	Country; Region; Youth NEET Rate; Incidence of tuberculosis	Grouped bar chart	Tuberculosis incidence increased consistently with higher youth NEET rates across all three regions, supporting the association between educational disengagement and greater disease burden.
11	Education reduces disease burden.	How does tertiary school life expectancy relate to TB incidence, controlling for access to electricity	Predictive	Country; Access to electricity; School life expectancy, tertiary; Incidence of tuberculosis	Scatter plot (color-grade for electricity) with trend	Higher tertiary school life expectancy was associated with lower tuberculosis incidence, with countries lacking electricity access showing the highest disease burden.
12	Education reduces disease burden.	How has tuberculosis incidence trended over time across Bachelor's attainment bands, and how does it relate to hypertension prevalence?	Descriptive	Country; Year; Bachelor's attainment; Incidence of tuberculosis; Prevalence of hypertension	Dual-axis stacked bar and line chart	TB incidence declined across all Bachelor's attainment bands from 2012 to 2019, while hypertension prevalence rose steadily, suggesting education reduced infectious but not chronic disease burden.

13	Education is associated with higher health expenditure and lifestyle-related health risks	Does the prevalence of overweight adults differ by compulsory education bands, and does controlling for GDP per capita modify the relationship?	Descriptive	Country; GDP per capita; Compulsory education duration; Prevalence of overweight	Grouped bar chart	Overweight prevalence increased with longer compulsory education duration across all GDP per capita bands, suggesting education alone is insufficient to address lifestyle-related health risks.
14	Education is associated with higher health expenditure and lifestyle-related health risks.	How does Bachelor's attainment relate to health expenditure by GDP per capita bands?	Descriptive	Country; GDP per capita; Educational attainment, at least Bachelor's or equivalent; Health expenditure per capita	Side-by-side bar chart	Health expenditure per capita was primarily driven by GDP per capita rather than Bachelor's attainment, with high GDP countries consistently recording the highest spending across all attainment bands.
15	Education is associated with higher health expenditure and lifestyle-related health risks.	How has the prevalence of overweight adults trended by tertiary school life expectancy band?	Descriptive	Country; Year; School life expectancy, tertiary; Prevalence of overweight	Multi-line chart	Overweight prevalence was consistently higher in mid and high tertiary school life expectancy countries, suggesting greater educational engagement was associated with increased lifestyle-related

						health risks over time.
16	Education is associated with higher health expenditure and lifestyle-related health risks.	How do health expenditure and Bachelor's attainment compare across regions?	Descriptive	Country; Region; Educational attainment, at least Bachelor's or equivalent; Health expenditure per capita	Dual-axis line over bar chart	Health expenditure and Bachelor's attainment declined in parallel across regions, with North America leading on both measures and Sub-Saharan Africa ranking lowest, confirming that higher educational attainment is associated with greater healthcare spending.
17	Education access is associated with stronger public health infrastructure and outcomes.	How does tertiary enrolment relate to DPT immunization, weighted by population?	Descriptive	Country; Total population; Enrolment in tertiary education; Immunization DPT	Bubble chart	Higher tertiary enrollment was generally associated with stronger DPT immunization coverage, though large-population countries showed that enrollment gains alone do not guarantee improved immunization outcomes.

18	Education access is associated with stronger public health infrastructure and outcomes.	How does hypertension vary by region and compulsory education duration?	Descriptive	Country; Region; Compulsory education duration; Prevalence of hypertension	Tree map	Hypertension prevalence was highest in Europe & Central Asia and Latin America & Caribbean regardless of compulsory education duration, indicating that regional demographic factors play a greater role in hypertension outcomes than mandatory schooling alone.
19	Education access is associated with stronger public health infrastructure and outcomes.	How does DPT immunization rate compare across tertiary school life expectancy bands	Descriptive	Country; School life expectancy, tertiary; Immunization DPT	Multi-box plot	As tertiary school life expectancy increased, DPT immunization rates rose and distributions narrowed, with the low band showing the greatest variability and lowest median, supporting the association between tertiary education access and stronger preventive healthcare outcomes.

20	Education access is associated with stronger public health infrastructure and outcomes.	How does health expenditure vary by region and tertiary school life expectancy?	Descriptive	Country; Region; School life expectancy, tertiary; Health expenditure per capita	Stacked bar chart	Regions with greater tertiary school life expectancy contributed the largest share of health expenditure, with North America and Europe & Central Asia leading across all bands, supporting the association between tertiary education access and stronger public health investment.
21	Statistical: Education indicators jointly predict crude death rate after controlling for economic factors.	Which education indicators most strongly explain crude death rate, controlling for GDP per capita?	Descriptive statistics + Multiple linear regression	Country; GDP per capita; Educational attainment, at least Bachelor's or equivalent; Youth NEET rate; School life expectancy, tertiary; Enrolment in tertiary education; Death rate, crude	Regression coefficient box plot	Youth NEET is the strongest, most consistent predictor of crude death rate across income levels, though which education indicators matter depends heavily on a country's GDP tier.

22	Machine learning: Countries cluster into distinct education-health profiles.	Can countries be grouped into interpretable clusters using their combined education and health indicators?	K-means clustering with PCA visualization	(All 5 education and all 7 health variables): Educational attainment, at least Bachelor's or equivalent; Compulsory education duration; Youth NEET rate; School life expectancy, tertiary; Enrolment in tertiary education; Death rate, crude; Fertility rate, total; Immunization DPT; Incidence of tuberculosis; Health expenditure per capita; Prevalence of overweight; Prevalence of hypertension	Cluster scatter plot (PCA reduced)	Yes, clustering on education and health indicators alone recovers four recognizable global development tiers without any predefined labels.

2. Visualizations and Insight

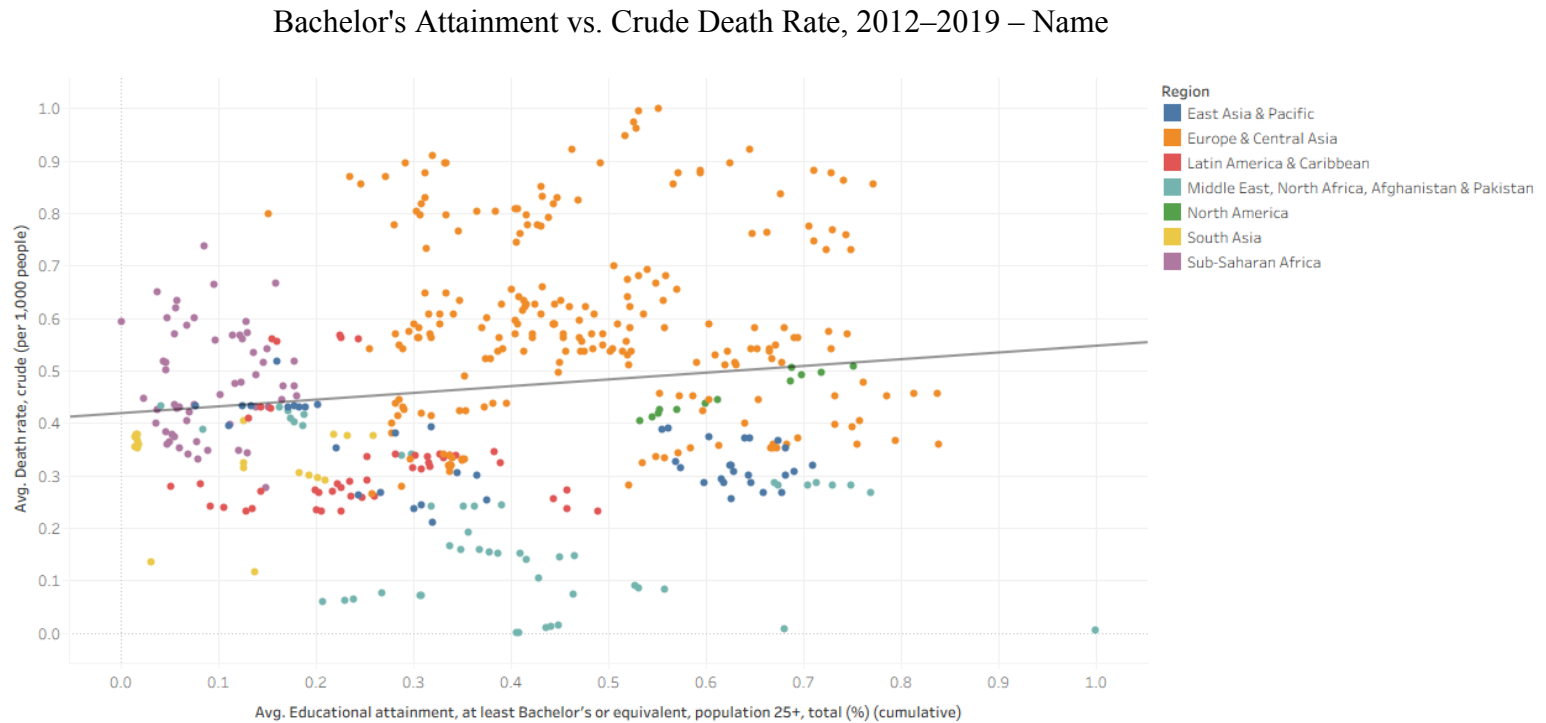


Figure 1: Scatter plot of average Bachelor's attainment (population 25+) vs. average crude death rate across 112 countries for the period 2012–2019, controlling for region.

- The scatter plot displayed average Bachelor's attainment (x) against crude death rate per 1,000 people (y) across 112 countries, color-coded by region.
- A weak positive trend ($r = 0.13$) was observed, with Europe and Central Asia driving higher death rates likely due to aging population structures.
- Crude death rate should be interpreted cautiously in high-attainment countries, as demographic aging may confound the education-mortality relationship.

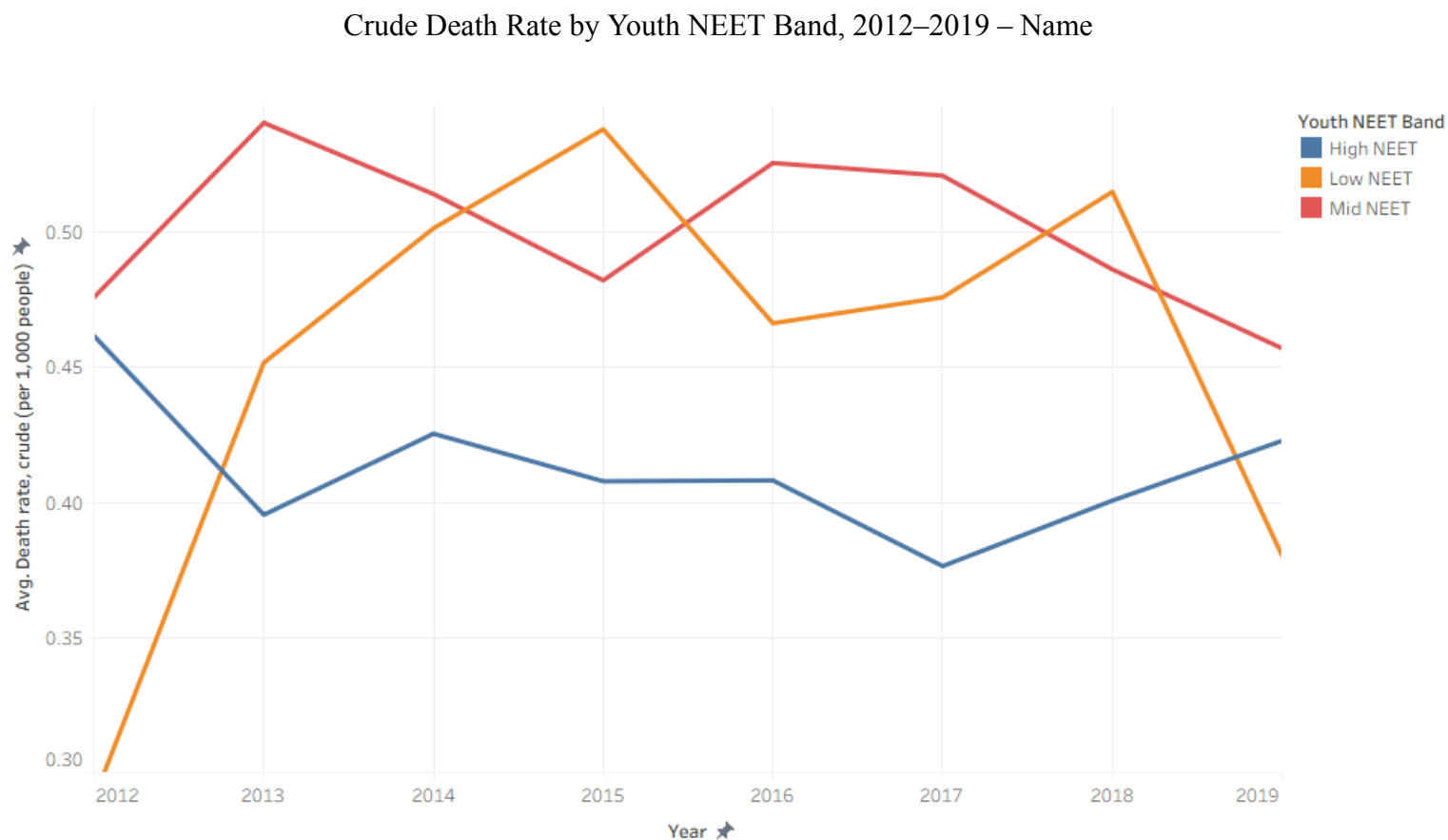


Figure 2: Multi-line chart of average crude death rate by youth NEET band across 112 countries for the period 2012–2019.

- The chart tracked average crude death rate (y) over time (x) across three youth NEET bands: low, mid, and high.
- Low and mid NEET countries showed higher death rates than high NEET countries, with all three bands converging by 2019.
- Countries should not rely solely on reducing NEET rates as a mortality reduction strategy, as death rate trends appeared largely independent of educational engagement levels over time.

Crude Death Rate Distribution by Compulsory Education Duration Band, 2012–2019 – Name

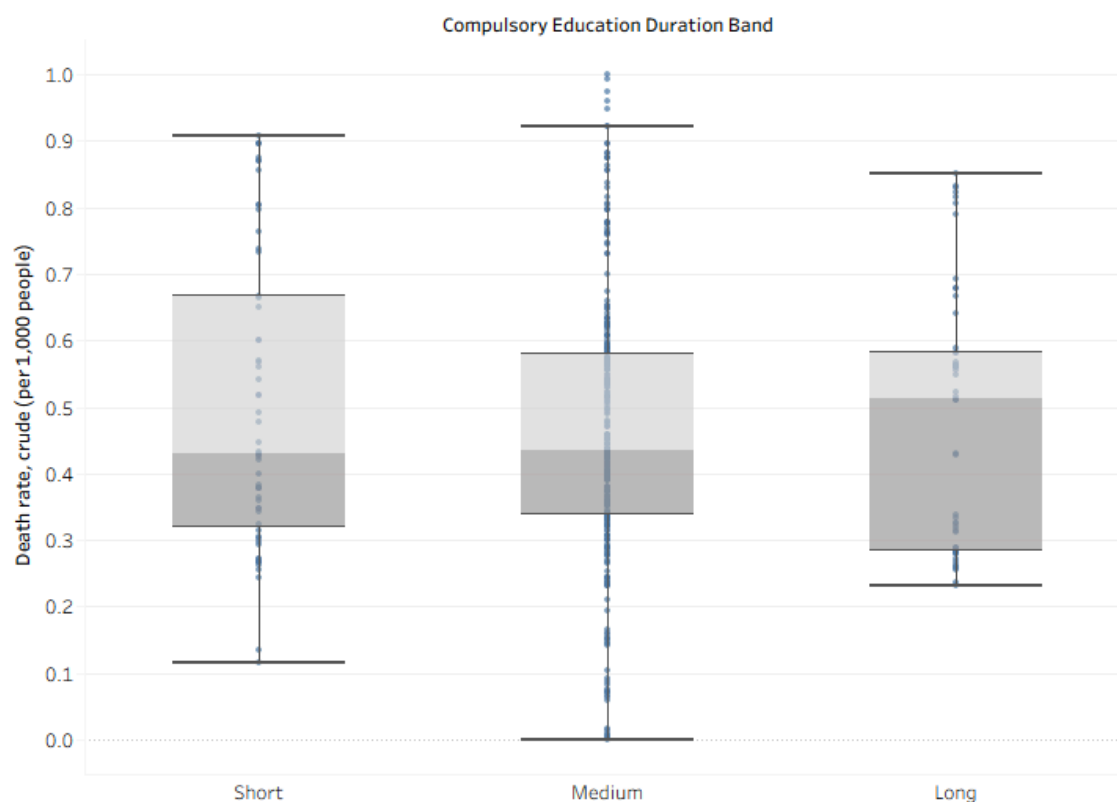


Figure 3: Multi box plot of death rate distribution by compulsory education duration band across 112 counties for the period 2012-2019.

- The box plot displayed the distribution of crude death rate per 1,000 people (y) across three compulsory education duration bands: short, medium, and long (x).
- Median death rates remained similar across all three bands, with the long duration band showing a slightly higher median and wider interquartile range than the short and medium bands.
- Extending compulsory education duration alone did not appear sufficient to reduce mortality, suggesting that education quality and broader health system factors play a more significant role.

Tertiary School Life Expectancy vs. Crude Death Rate by Population, 2012-2019 – Name

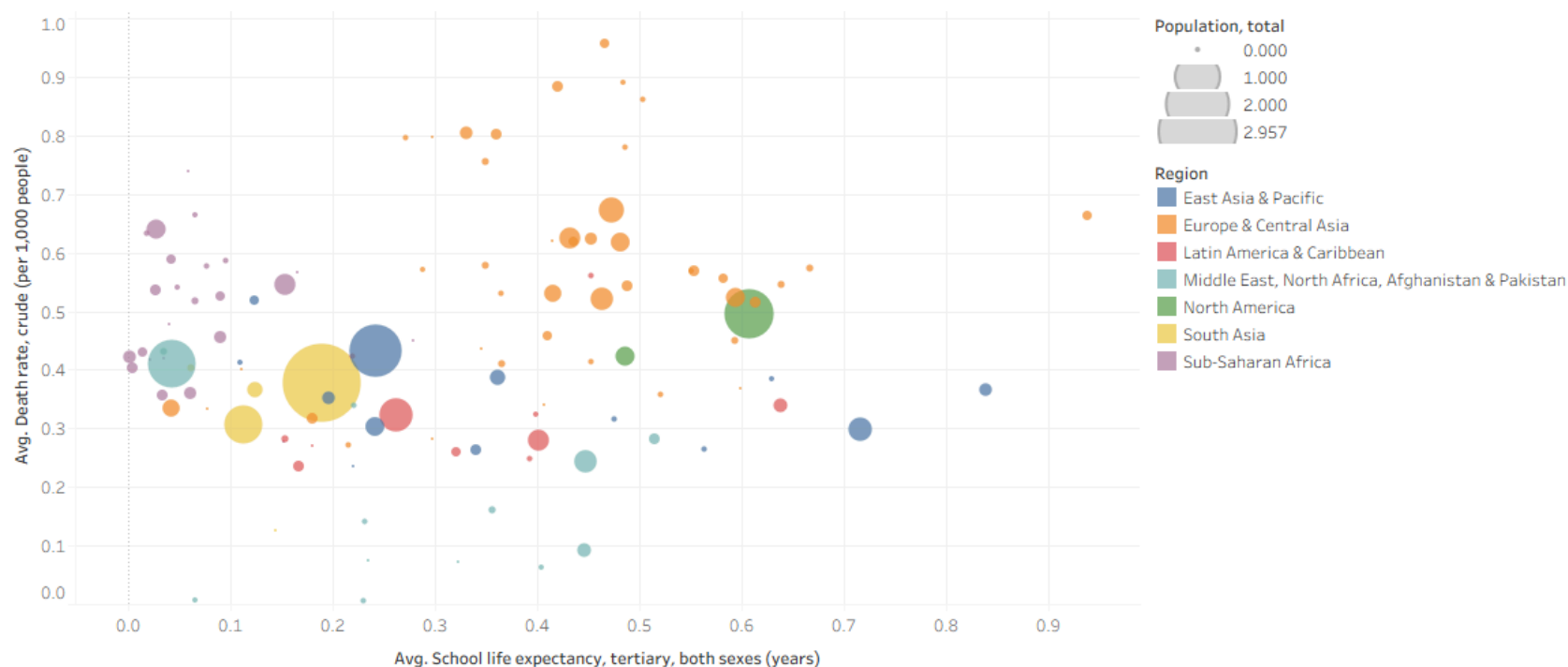


Figure 4: Bubble chart of average tertiary school life expectancy vs. Average crude death rate across 112 countries for the period 2012-2019, controlling for Population and Region.

- The bubble chart showcased average tertiary school life expectancy (x) against average crude death rate per 1,000 people (y), with bubble size representing total population and color indicating region.
- A weak positive association ($r = 0.21$) was observed, with larger-population countries in South Asia and East Asia clustering at mid-range school life expectancy and moderate death rates.
- Crude death rate is not a reliable measure of health quality in highly educated countries due to aging populations. Policymakers should focus on age-adjusted mortality indicators for a more accurate assessment.

Bachelor's Attainment vs. Fertility Rate by GDP Per Capita Band, 2012–2019 – Name



Figure 5: Multi-scatter plot of average Bachelor's attainment vs. average fertility rate across 112 countries for the period 2012–2019, controlling for GDP per capita.

- Average Bachelor's attainment (x) was plotted against average fertility rate in births per woman (y), split across three GDP per capita bands: low, mid, and high.
- A strong negative association was observed in low GDP countries ($r = -0.52$), with higher attainment linked to significantly lower fertility rates, while mid and high GDP bands showed a flatter, near-zero relationship.
- Policymakers in low-income countries should prioritize Bachelor's attainment expansion as a key strategy for reducing fertility rates, as the association was strongest where educational access remains limited.

DPT Immunization Coverage by Tertiary School Life Expectancy Band, 2012–2019 – Name

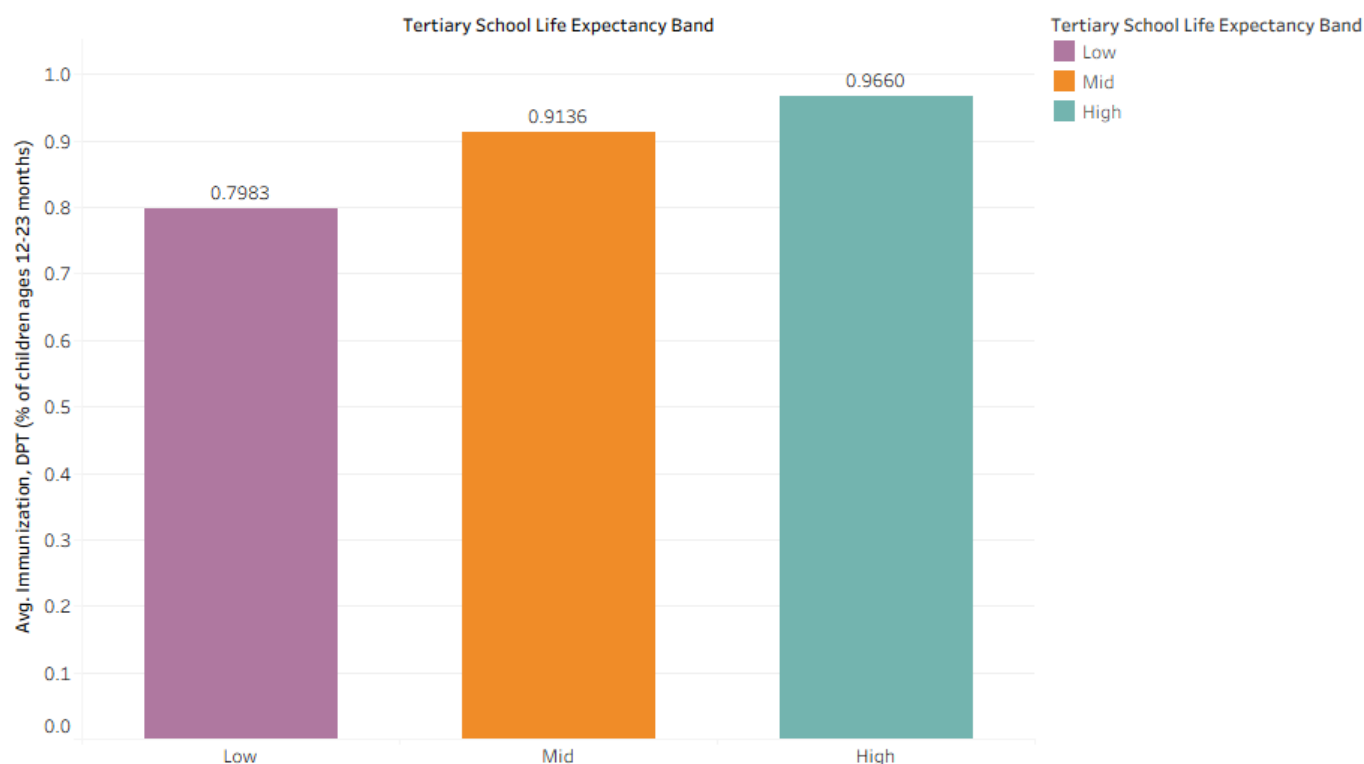


Figure 6: Bar chart of average DPT immunization coverage by tertiary school life expectancy band across 112 countries for the period 2012–2019

- Average DPT immunization coverage (y) was compared across three tertiary school life expectancy bands: low (0.80), mid (0.91), and high (0.97) (x).
- A positive pattern was observed, with countries in the high tertiary school life expectancy band achieving nearly universal DPT immunization coverage compared to those on the low band.
- Expanding tertiary education access should be considered an effective strategy for improving childhood immunization rates, particularly in countries with low school life expectancy.

Fertility Rate by Bachelor's Attainment Band Over Time, 2012-2019 – Name

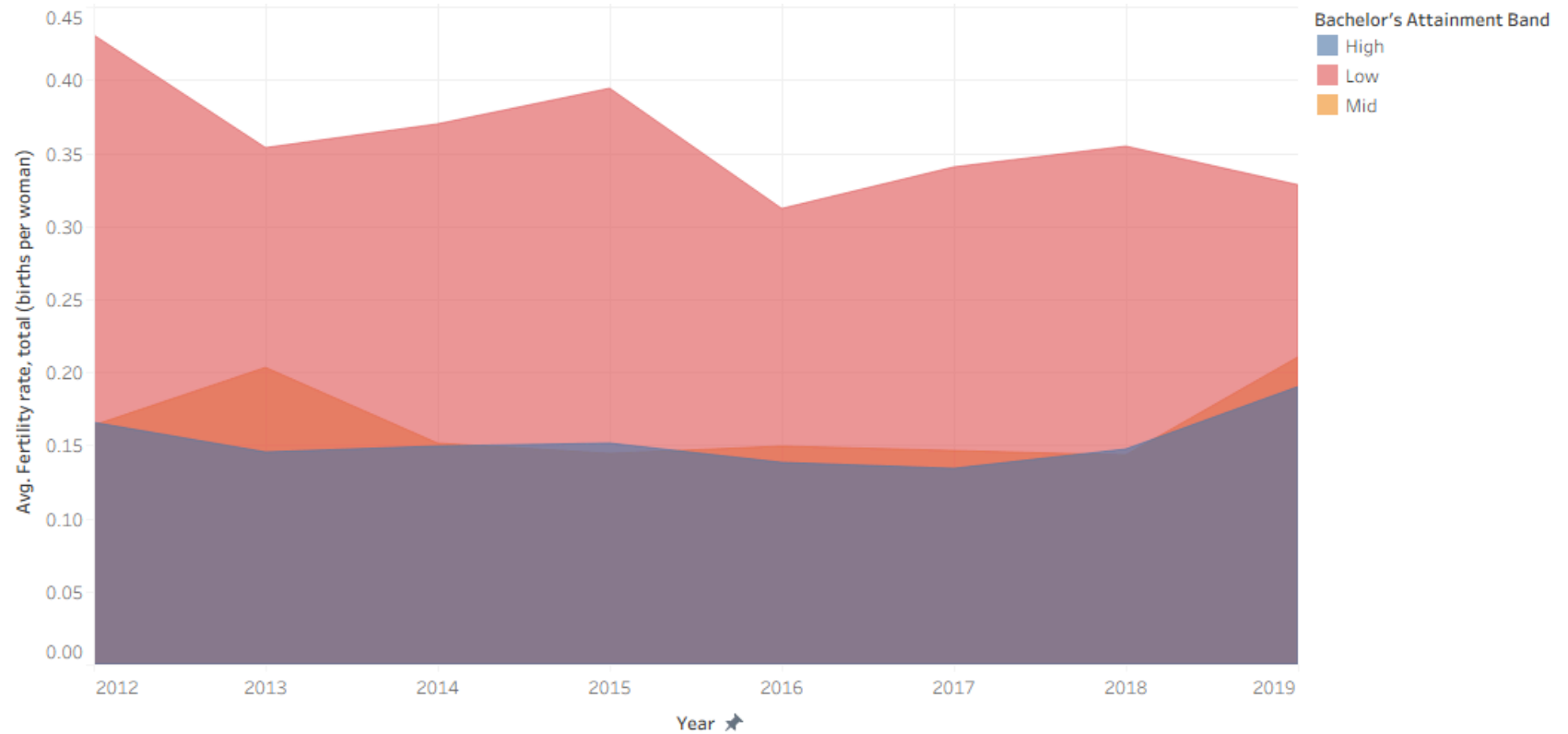


Figure 7: Multi-line area chart of average fertility rate by Bachelor's attainment band across 112 countries for the period 2012–2019.

- Average fertility rate (y) was tracked over time (x) across three Bachelor's attainment bands: low, mid, and high.
- Low attainment countries consistently accounted for the largest share of fertility rate throughout the period, while high attainment countries maintained the lowest and most stable fertility rates over time.

- Policymakers should prioritize Bachelor's attainment expansion as a key strategy for reducing fertility rates over time, particularly in low attainment countries where the gap remains widest.

Compulsory Education Duration vs. Fertility Rate, 2012–2019 – Name

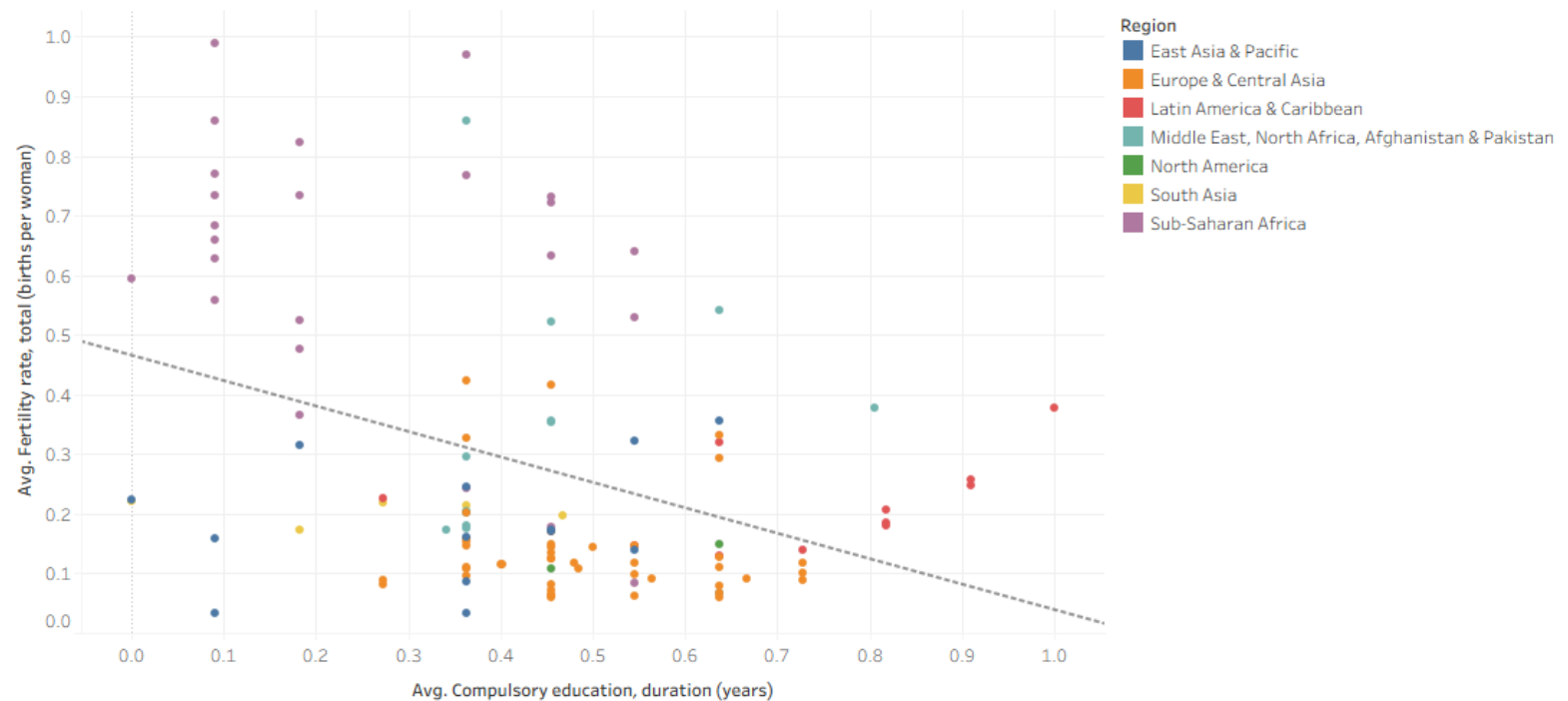


Figure 8: Scatter plot of average compulsory education duration vs. average fertility rate across 112 countries for the period 2012–2019, controlling for region.

- Average compulsory education duration (x) was plotted against average fertility rate in births per woman (y), color-coded by region.
- A negative trend was observed, with higher compulsory education duration associated with lower fertility rates, particularly among European and Central Asian countries.

- Policymakers in high-fertility regions should consider extending compulsory education duration as an effective tool for reducing fertility rates at the country level.

TB Incidence by Region and Bachelor's Attainment Band, 2012–2019

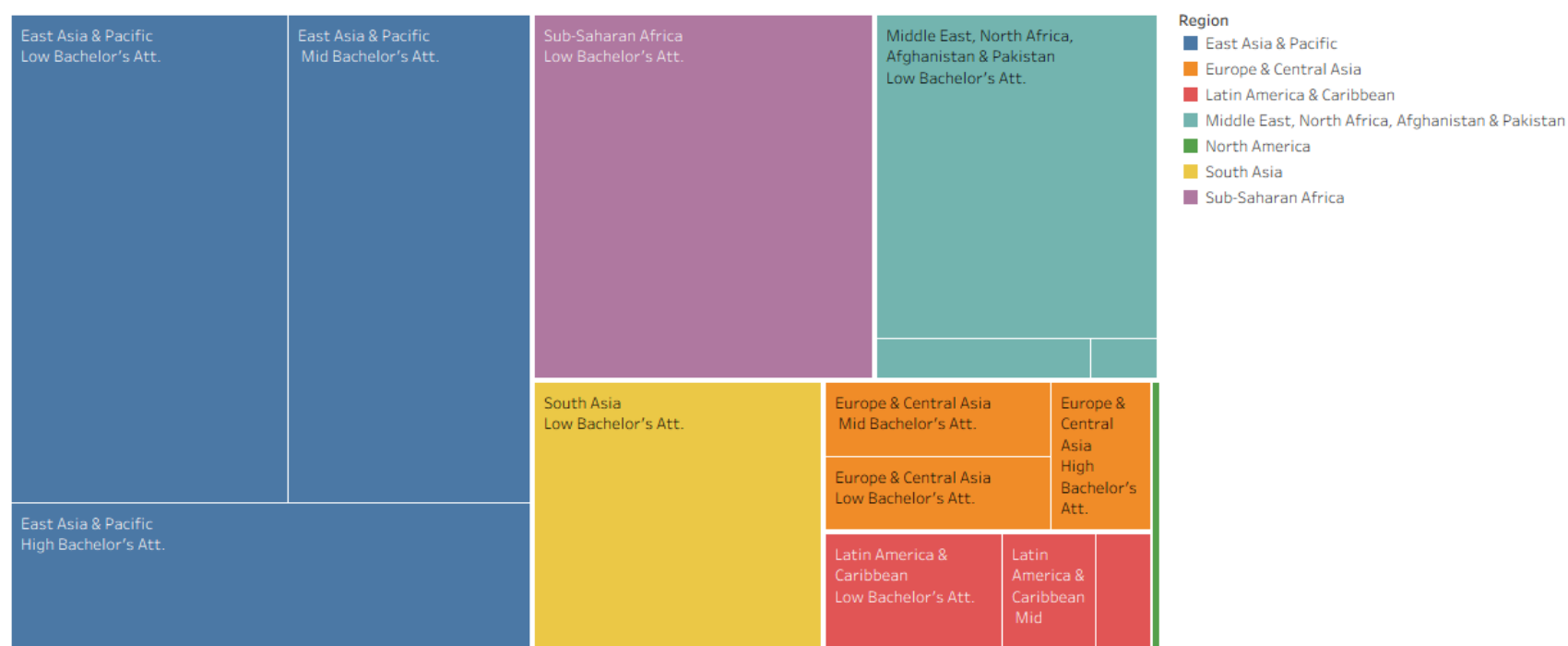


Figure 9: Tree map where block size reflects average TB incidence by region and Bachelor's attainment band across 112 countries for the period 2012–2019

- Each block represents a region and Bachelor's attainment band combination, with block size indicating average TB incidence and color representing region.

- Sub-Saharan Africa and South Asia dominate the tree map at low Bachelor's attainment levels, while Europe & Central Asia blocks shrink considerably at higher attainment bands, indicating that higher educational attainment is associated with lower TB burden.
- Policymakers should prioritize expanding Bachelor's level education in high TB-burden regions as a long-term strategy for reducing infectious disease prevalence.

TB Incidence by youth NEET Band Across Selected Regions, 2012-2019 – Name

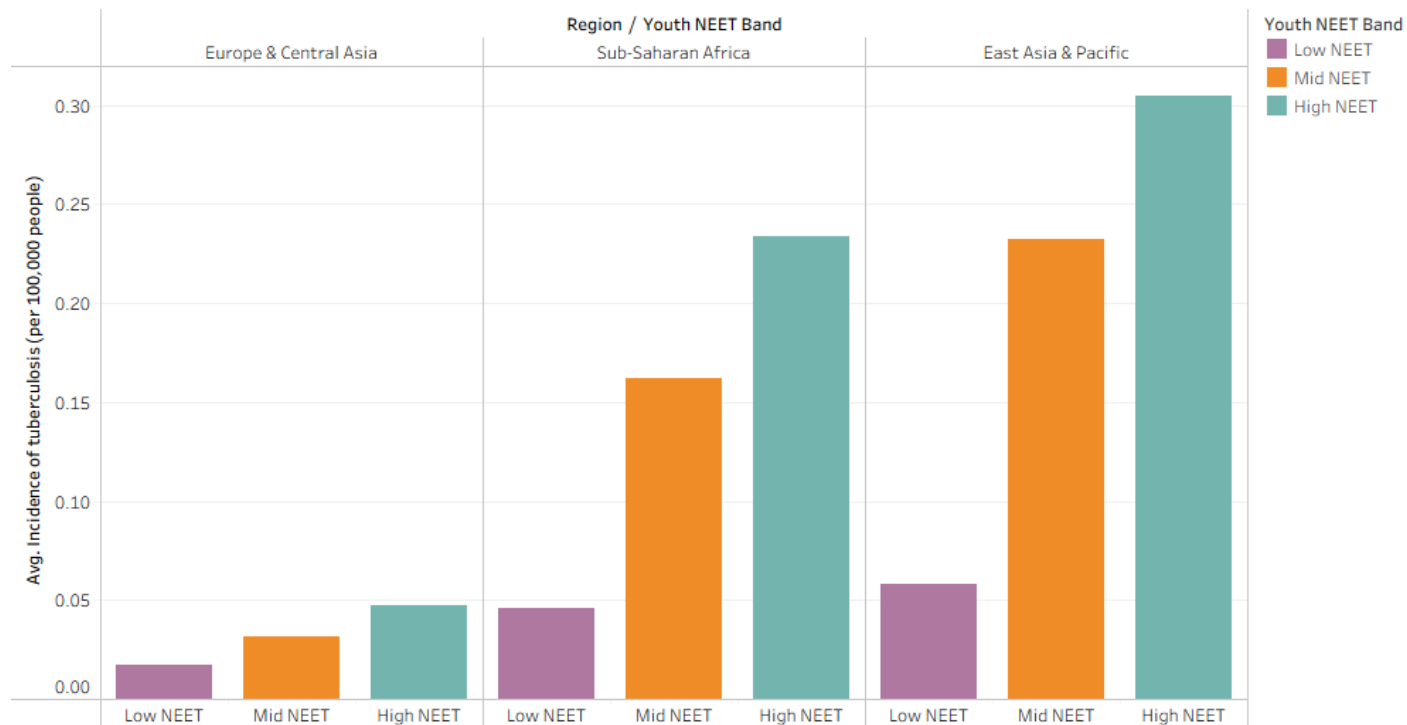


Figure 10: Grouped bar chart of average tuberculosis incidence by youth NEET band across three selected regions (Europe and Central Asia, Sub-Saharan Africa, and East Asia and Pacific) for the period 2012–2019, controlling for region.

- Average tuberculosis incidence per 100,000 people (y) was compared across low, mid, and high youth NEET bands (x) for three selected regions.
- Higher youth NEET bands were consistently associated with greater tuberculosis incidence across all three regions, with Sub-Saharan Africa and East Asia and Pacific showing the steepest increases from low to high NEET.
- Reducing youth NEET rates should be treated as a public health priority, as greater educational disengagement was linked to higher tuberculosis burden across all regions examined.

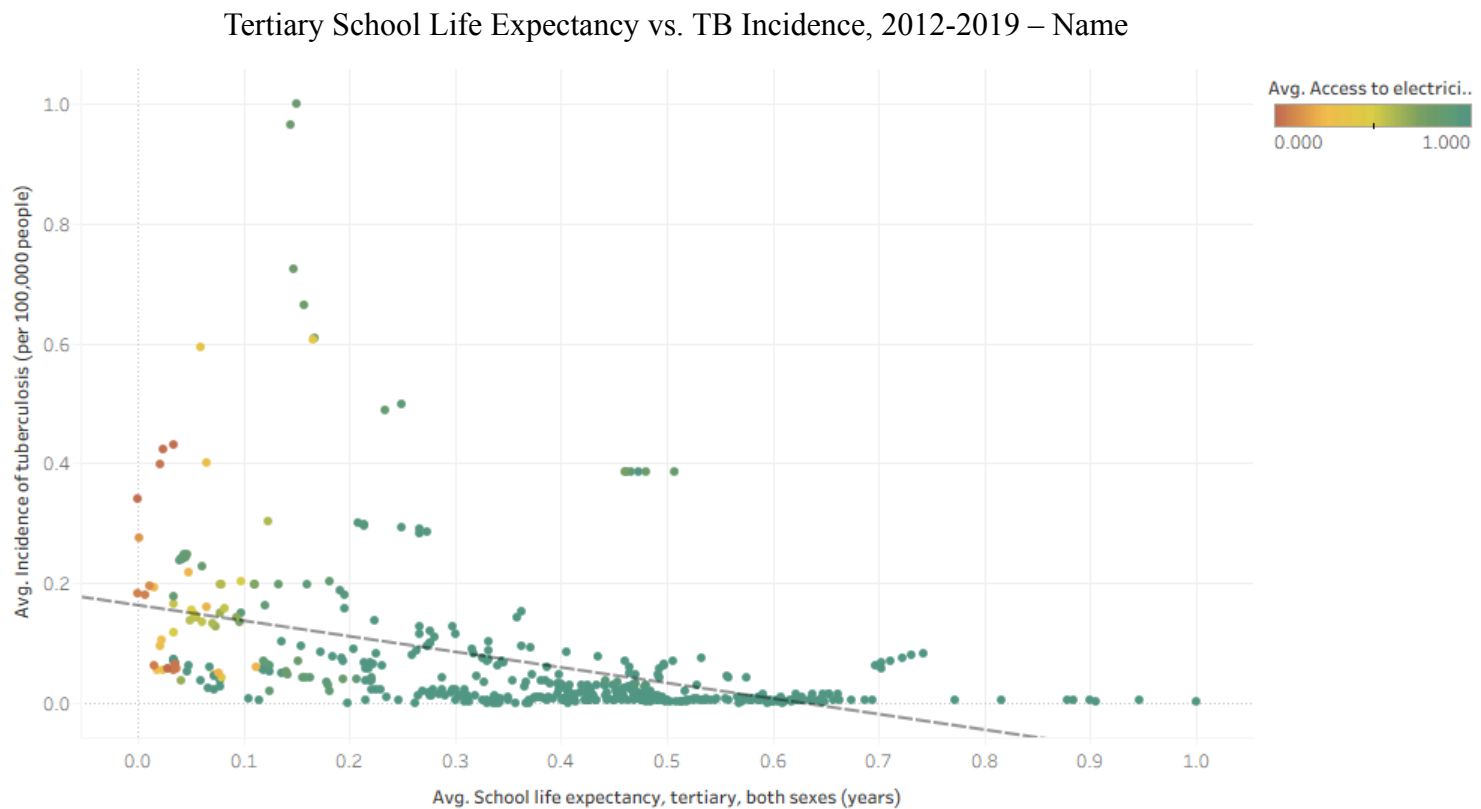


Figure 11: Scatter plot of average tertiary school life expectancy vs. average tuberculosis incidence across 112 countries for the period 2012–2019, controlling for access to electricity.

- Average tertiary school life expectancy (x) was plotted against average tuberculosis incidence per 100,000 people (y), with color gradient representing access to electricity.
- A moderate negative association ($r = -0.42$) was observed, with countries with higher tertiary school life expectancy showing lower TB incidence; countries with low electricity access clustered at low school life expectancy and high TB incidence.
- Policymakers should invest in tertiary education and electricity infrastructure development, as limited electricity access appeared to compound the disease burden in low school life expectancy countries.

TB Incidence by Bachelor's Attainment Band and Hypertension Prevalence Over Time, 2012–2019 – Name

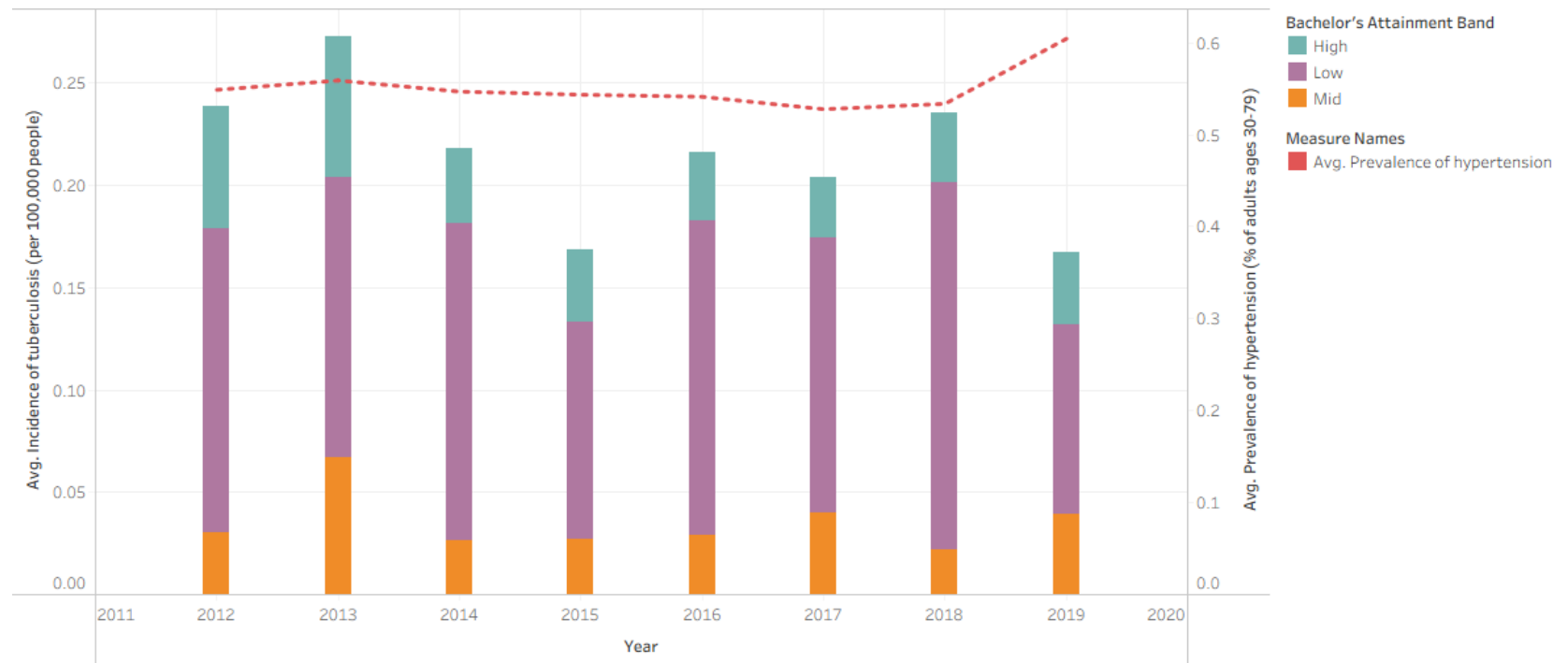


Figure 12: Dual-axis stacked bar and line chart of average tuberculosis incidence by Bachelor's attainment band and average hypertension prevalence across 112 countries for the period 2012–2019.

- Average tuberculosis incidence per 100,000 people (y-left) was tracked over time (x) across three Bachelor's attainment bands: low, mid, and high, with average hypertension prevalence (y-right) overlaid as a dashed line.
- TB incidence declined from 2012 to 2019 across all attainment bands, while hypertension prevalence rose steadily throughout the same period, suggesting education may reduce disease burden but has limited effect on chronic disease outcomes.
- Policymakers should implement targeted chronic disease interventions alongside education investment, as rising hypertension prevalence indicated that education alone was insufficient to address disease burden.

Overweight Prevalence by Compulsory Education Duration and GDP Per Capita Band, 2012–2019 – Name

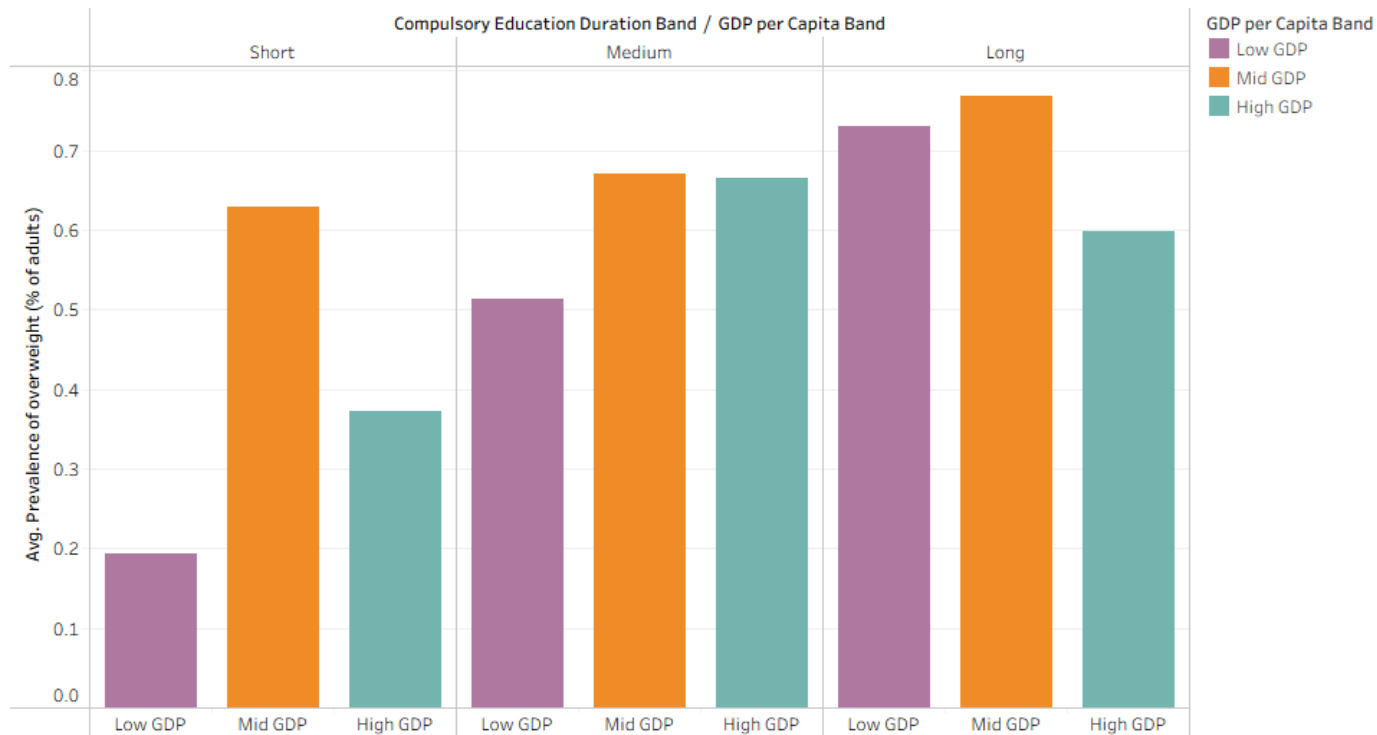


Figure 13: Grouped bar chart of average overweight prevalence by compulsory education duration band across 112 countries for the period 2012–2019, controlling for GDP per capita.

- Average overweight prevalence (y) was compared across three compulsory education duration bands: short, medium, and long (x), with each band further split by GDP per capita band.
- A positive relationship between compulsory education duration and overweight prevalence was observed across all GDP bands, most notably among low GDP countries moving from short to long duration, suggesting that longer schooling coincides with lifestyle changes associated with higher overweight rates.
- Policymakers should incorporate nutrition and physical activity programs into education, as longer compulsory education duration appeared associated with greater overweight prevalence regardless of income level.

Bachelor's Attainment vs. Health Expenditure Per Capita by GDP Per Capita Band, 2012–2019 – Name

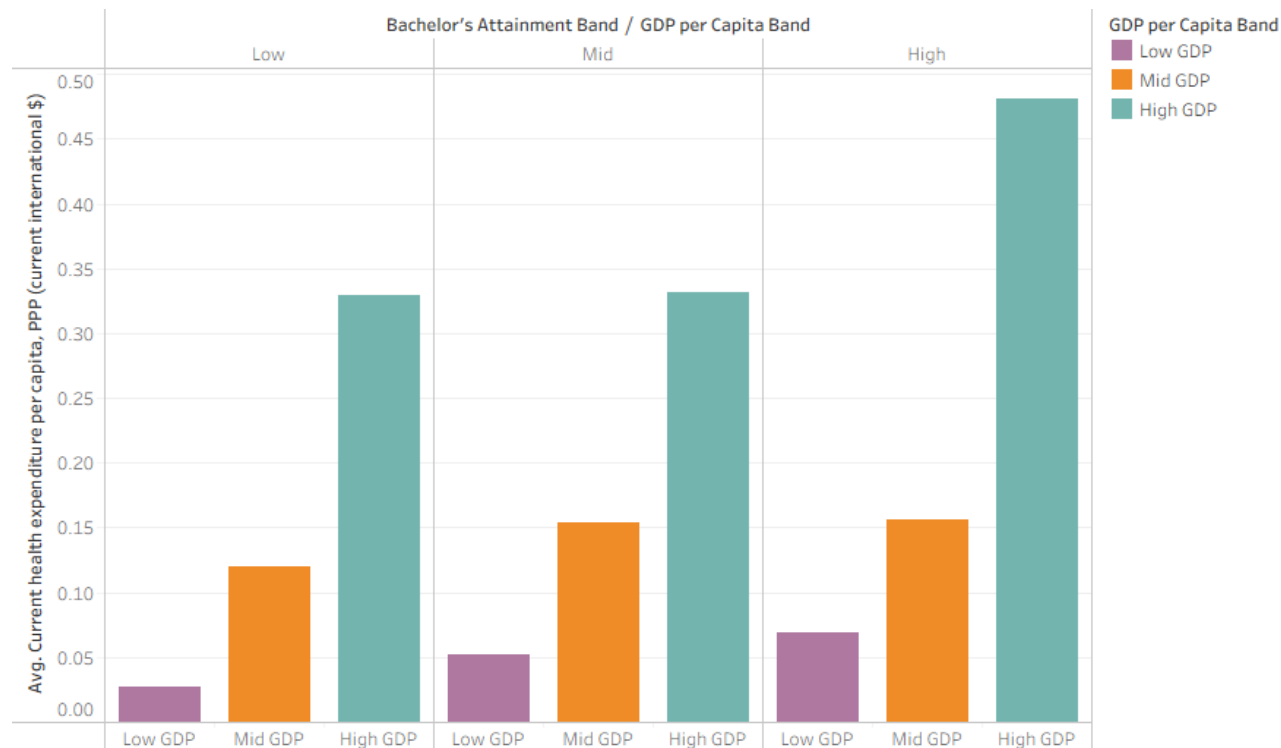


Figure 14: Side-by-side bar chart of average health expenditure per capita by Bachelor's attainment band across 112 countries for the period 2012–2019, controlling for GDP per capita.

- Average health expenditure per capita in PPP (y) was compared across three Bachelor's attainment bands: low, mid, and high (x), with each band further split by GDP per capita band.
- Health expenditure increased consistently with higher GDP per capita across all attainment bands, with high GDP countries spending significantly more regardless of attainment level, suggesting GDP was a stronger driver of health expenditure than education alone.
- Policymakers should account for income level when assessing the education-health expenditure relationship, as economic capacity appeared to outweigh educational attainment as a determinant of healthcare spending.

Overweight Prevalence by Tertiary School Life Expectancy Band Over Time, 2012–2019 – Name

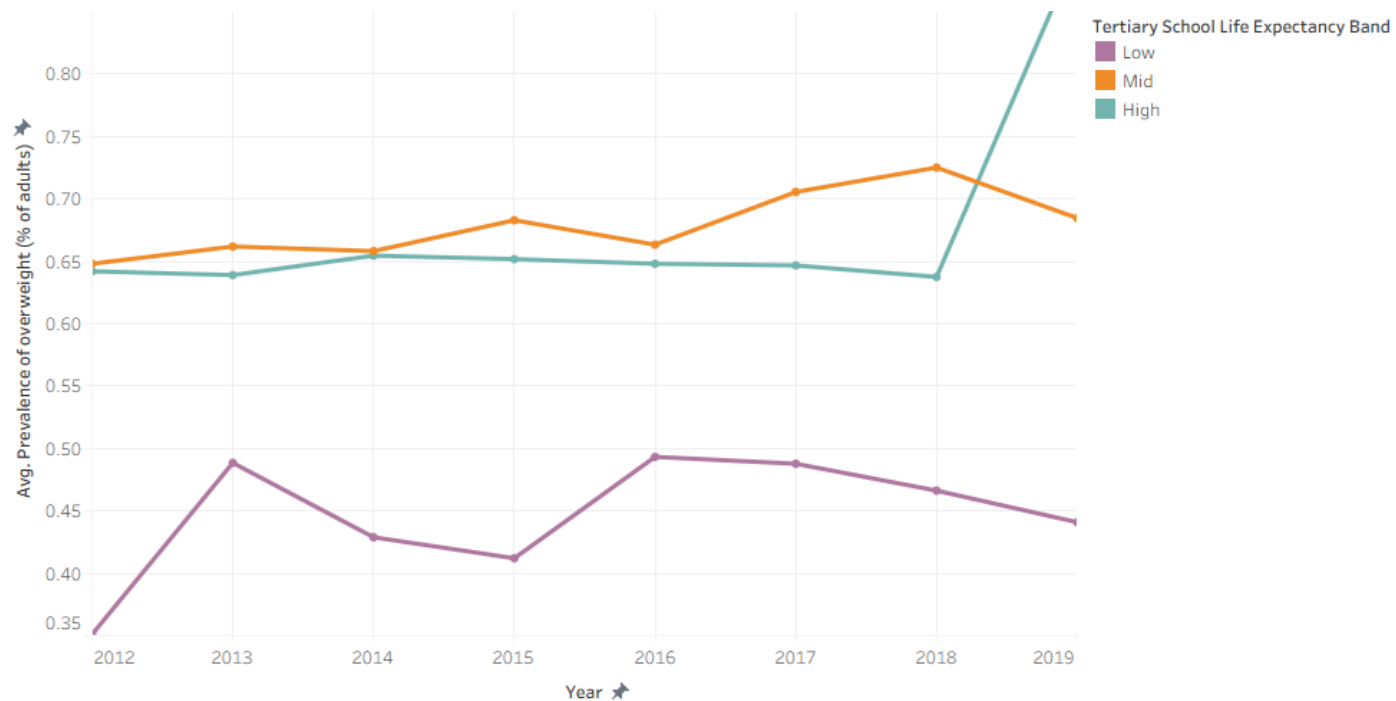


Figure 15: Multi-line chart of average overweight prevalence by tertiary school life expectancy band across 112 countries for the period 2012–2019

- Average overweight prevalence (y) was tracked over time (x) across three tertiary school life expectancy bands: low, mid, and high.
- Mid and high school life expectancy countries maintained consistently higher overweight prevalence throughout the period, with the high band showing a sharp increase in 2019, while low school life expectancy countries remained notably lower across all years.
- Policymakers should not rely on tertiary education expansion alone to address overweight prevalence, as higher educational engagement appeared associated with greater rather than lower lifestyle-related health risks.

Bachelor's Attainment and Health Expenditure by Region, 2012-2019 – Name

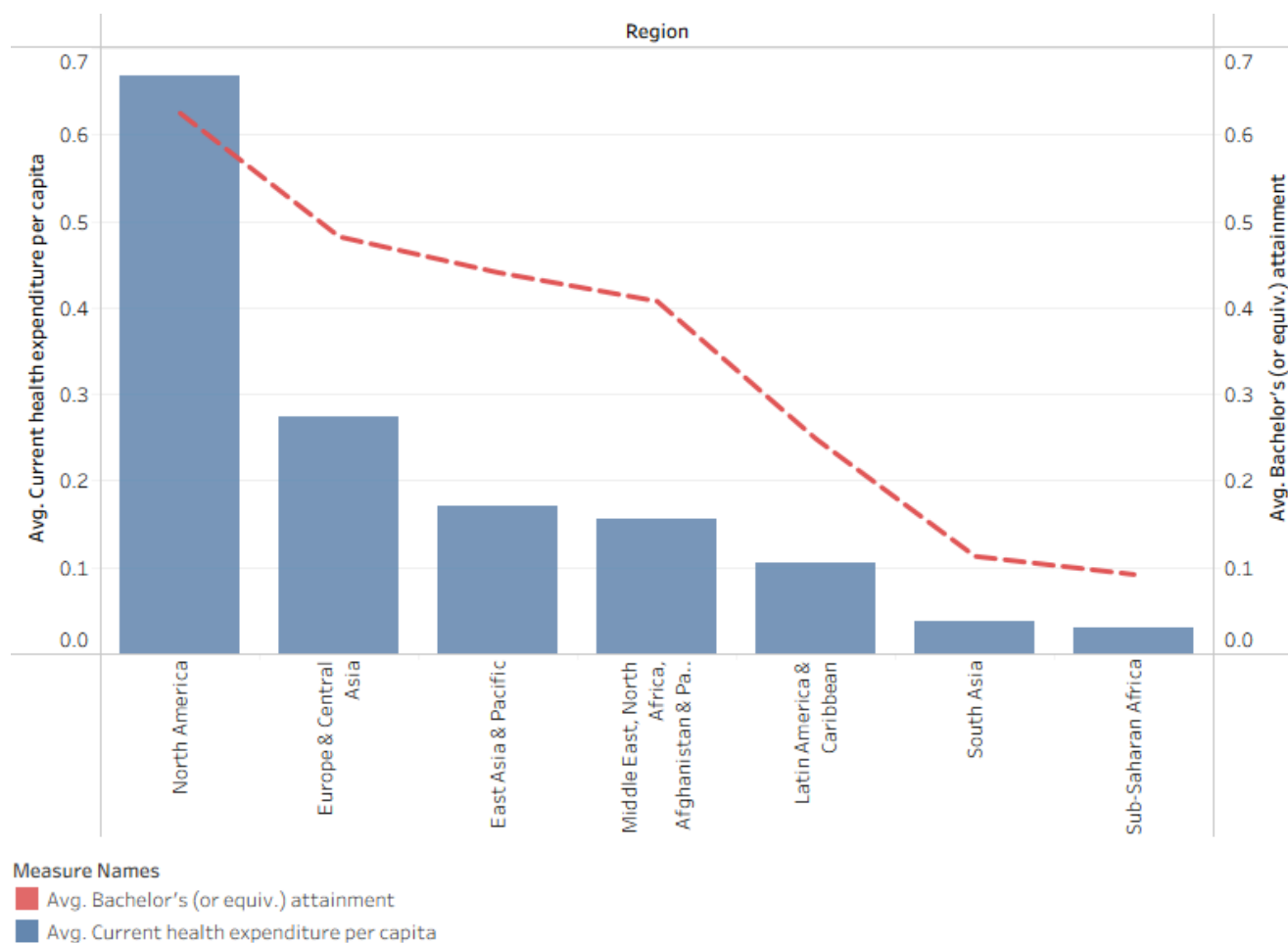


Figure 16: Dual-axis chart of average health expenditure per capita (bar) and average Bachelor's attainment (line) across seven World Bank regions for the period 2012-2019

- Average health expenditure per capita (left axis, bar) and average Bachelor's attainment (right axis, dashed line) are displayed across seven regions sorted from highest to lowest expenditure.
- Both measures decline in near-paralleled fashion from North America and Europe & Central Asia to South Asia and Sub-Saharan Africa, suggesting a consistent regional alignment between educational attainment and healthcare spending capacity.
- Policymakers in lower-income regions should prioritize higher education investment as a long-term strategy for increasing healthcare spending capacity, given the strong connection between Bachelor's attainment and health expenditure.

Tertiary Enrolment vs. DPT Immunization by Region and Population, 2012-2019 - Name

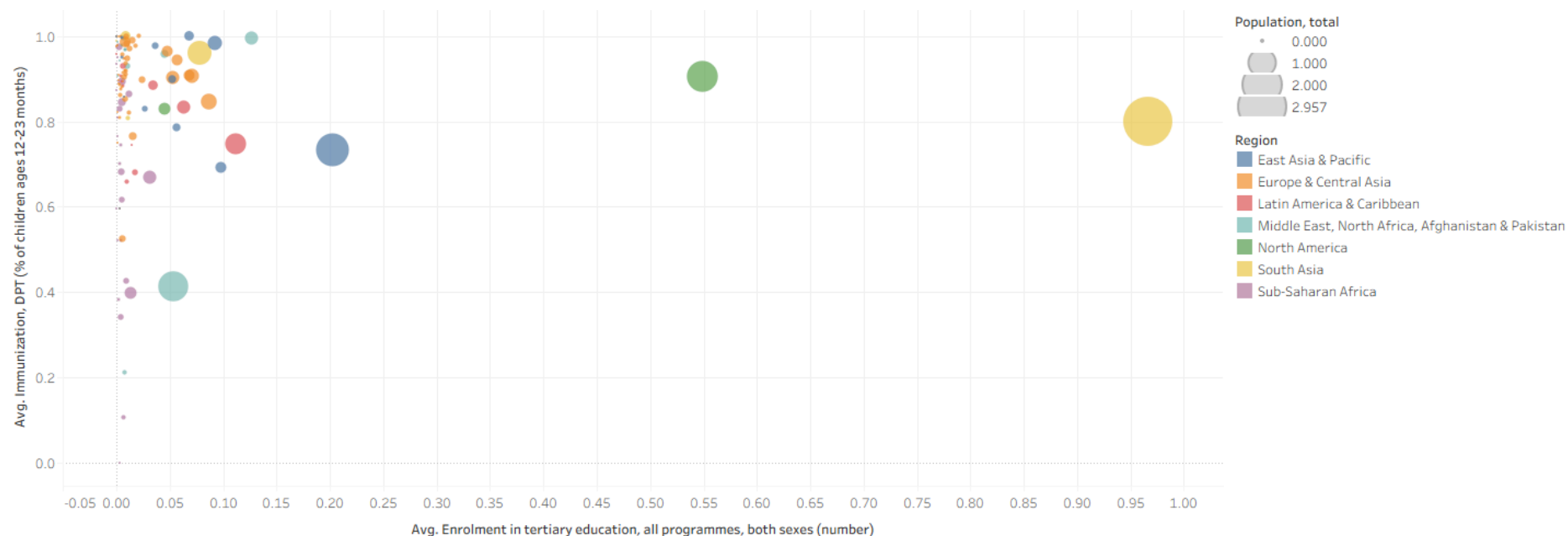


Figure 17: Bubble chart of average DPT Immunization rate by tertiary enrolment across 112 countries for the period 2012-2019, controlling for population and region.

- Average DPT Immunization (y) is plotted against tertiary enrolment (x) with bubble size representing population and color representing region.
- Most countries cluster at a low enrolment with high immunization rates, while the largest-population countries such as South Asia and East Asia & Pacific show wide enrolment variation, suggesting enrolment does not drive immunization outcomes.
- Tertiary education alone is insufficient to drive immunization coverage; complementary health system investments are needed to close the gap in underperforming regions.

Hypertension Prevalence by Region and Compulsory Education Duration Band, 2012-2019 – Name



Figure 18: Tree map of average hypertension prevalence by region and compulsory education duration band across 112 countries for the period 2012-2019, where block size indicates hypertension prevalence

- Hypertension prevalence is displayed across region and compulsory education duration band combinations, with block size indicating the magnitude of prevalence and color representing region.
- Europe & Central Asia and Latin America & Caribbean dominate the tree map regardless of compulsory education duration, suggesting that regional and demographic factors outweigh mandatory schooling duration as drivers of hypertension prevalence.

- Policymakers should look beyond compulsory education duration when designing hypertension reduction strategies, as region-specific factors such as aging populations and dietary patterns appear to be stronger determinants.

DPT Immunization Rate by Tertiary School Life Expectancy Band, 2012-2019 – Name

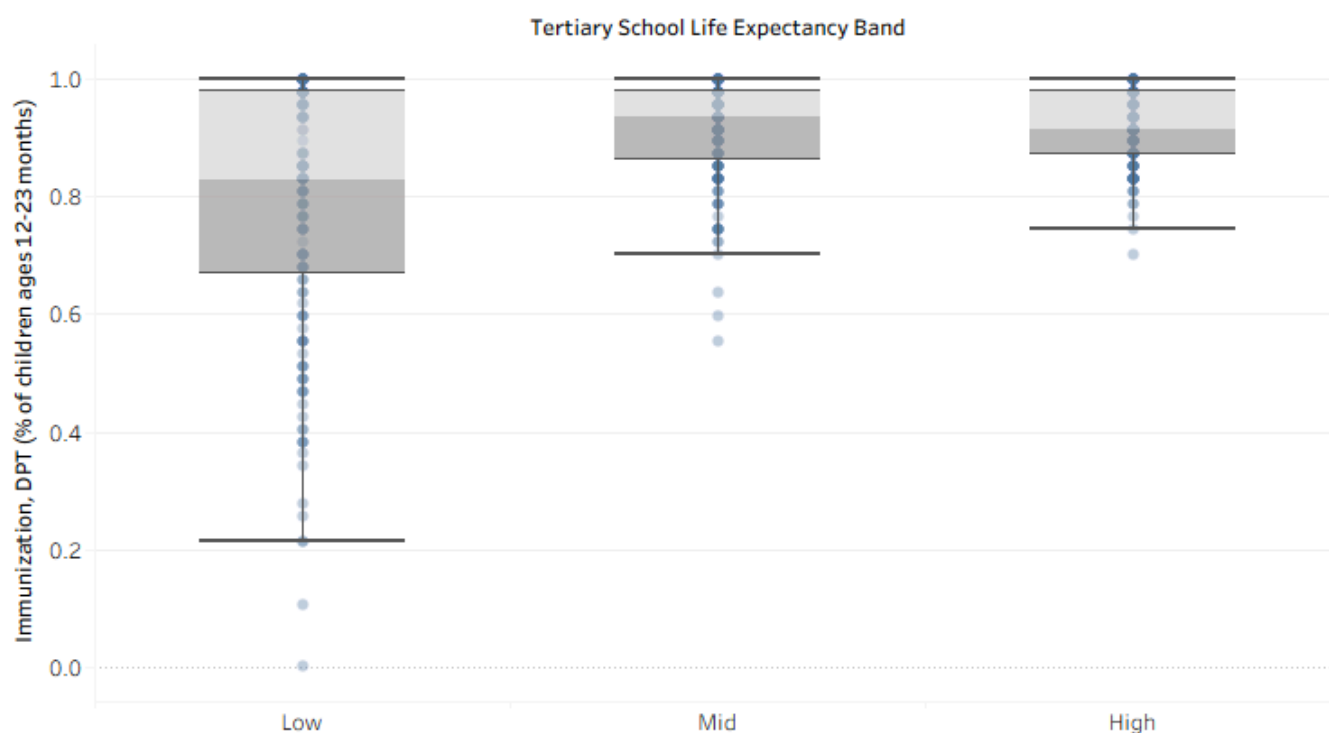


Figure 19: Box plot of average DPT Immunization rate by tertiary school life expectancy band across 112 countries for the period 2012-2019.

- DPT immunization distribution is compared across three tertiary school life expectancy bands (low, mid and high), with each box displaying the median, interquartile range and outliers.
- The low band shows the widest spread and lowest median immunization rate, while mid and high bands are tighter and higher, indicating that greater tertiary education access is associated with more consistent and higher immunization coverage.
- Policymakers should prioritize expanding tertiary education access in low school life expectancy countries as a pathway to improving both the level and consistency of child immunization rates.

Health Expenditure Per Capita by Region and Tertiary School Life Expectancy Band, 2012–2019

– Name

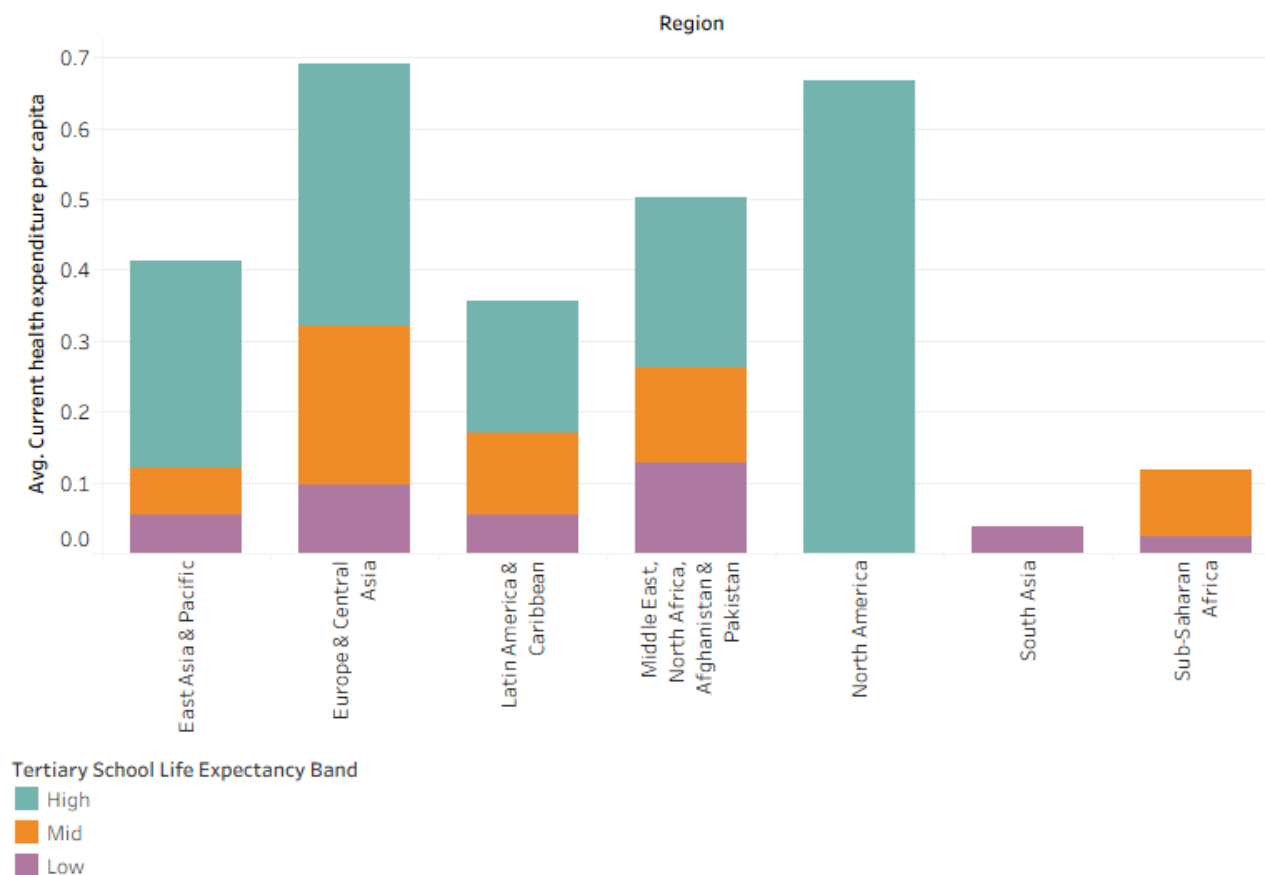


Figure 20: Stacked bar chart of average health expenditure per capita by region and tertiary school life expectancy band across 112 countries for the period 2012–2019

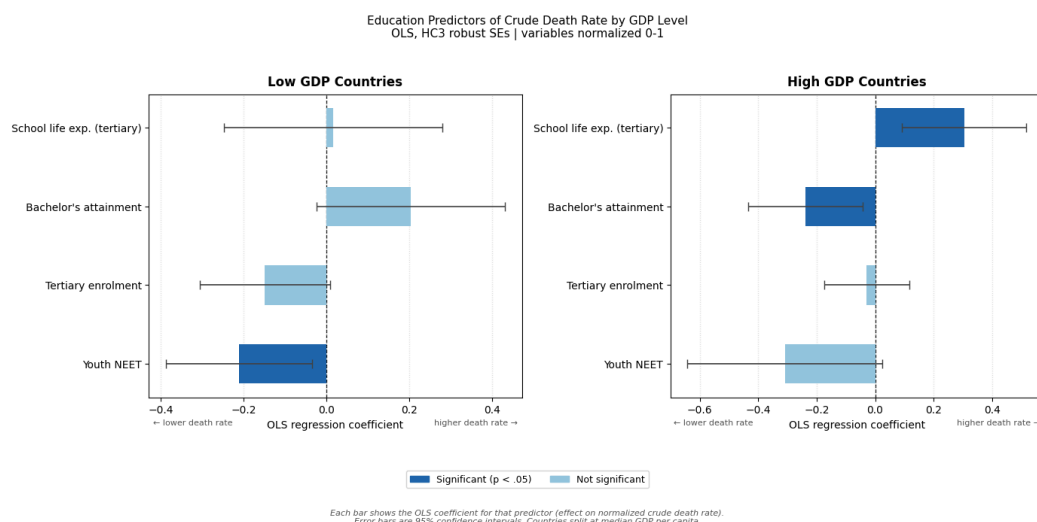
- Average health expenditure per capita (y) is broken down by tertiary school life expectancy band (low, mid, high) across seven regions, with each stacked segment representing a band's contribution to total expenditure.
- North America and Europe & Central Asia show the largest total expenditure with high school life expectancy band contributing to the greatest share, while South Asia and Sub-Saharan Africa remain consistently low across all bands.
- Policymakers in low-expenditure regions should invest in expanding tertiary education access as a means of driving long-term growth in healthcare spending capacity.

3. Statistical Model – Stratified Regression Analysis of Education, Income and Mortality

Hypothesis – Key education indicators have a statistically significant relationship with crude death rate, and that relationship differs between lower and higher-income countries.

Regression Results:

gdp_group	label	coefficient	ci_lower	ci_upper	p_value	significant
Low GDP	Bachelor's attainment	0.203771	-0.024140	0.431682	0.079711	False
Low GDP	Youth NEET	-0.210534	-0.387373	-0.033695	0.019627	True
Low GDP	School life exp. (tertiary)	0.016939	-0.247222	0.281100	0.899984	False
Low GDP	Tertiary enrolment	-0.148183	-0.305809	0.009443	0.065395	False
High GDP	Bachelor's attainment	-0.238911	-0.434322	-0.043500	0.016563	True
High GDP	Youth NEET	-0.308017	-0.641611	0.025578	0.070344	False
High GDP	School life exp. (tertiary)	0.304796	0.092010	0.517582	0.004993	True
High GDP	Tertiary enrolment	-0.028995	-0.174746	0.116757	0.696612	False



This dual box plot shows OLS coefficients for four education predictors of crude death rate, estimated separately within low-GDP per capita and high-GDP per capita country groups, with 95% confidence intervals and color indicating significance level.

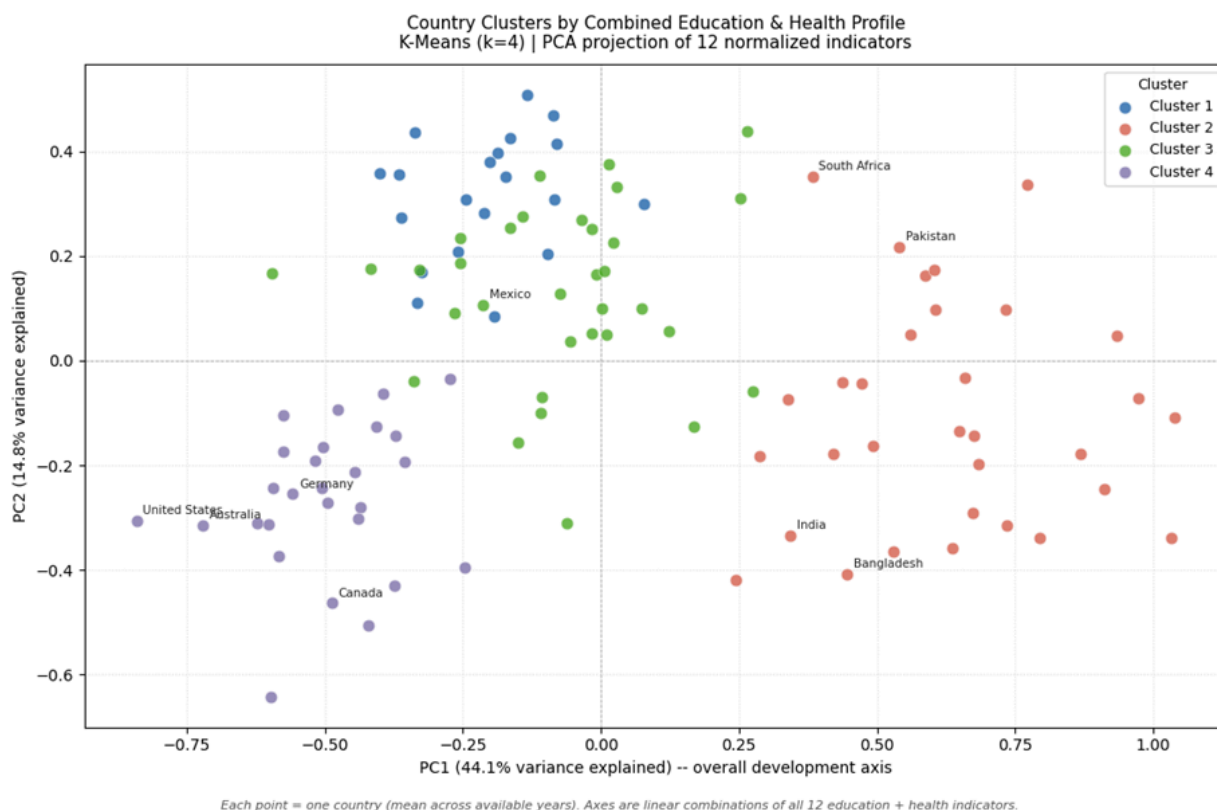
Youth NEET rate is the only predictor significant in both groups (negative in both), while Bachelor's attainment and school life expectancy flip direction, they are associated with higher death rates in low-GDP per capita countries but lower in high-GDP per capita ones.

The education-to-health relationship does not seem to be universal. Income context shapes which education factors matter, so policy recommendations should be tailored to a country's development level rather than applied globally.

4. ML Model – K-Means Clustering on Combined Education & Health Profiles

Hypothesis – Countries share enough structural similarity in their combined education and health profiles that unsupervised clustering can group them into meaningful development tiers without using any predefined labels.

Cluster Results:



This PCA scatter plot shows 112 countries colored by K-means cluster assignment ($k=4$), derived from 5 education indicators (bachelor's attainment, compulsory education duration, youth NEET, school life expectancy, tertiary enrolment) and 7 health indicators (crude death rate, fertility rate, DPT immunization, TB incidence, health expenditure, overweight prevalence, hypertension prevalence).

The four clusters recover recognizable global development tiers without being told what they are: high-income OECD countries (Norway, Canada), upper-middle transition economies (Croatia, Bulgaria), emerging middle-income countries (Thailand, Jordan), and lower-income/high-fertility countries (Tanzania, Rwanda). PC1, which captures 44% of variance, is dominated by a fertility-vs-education axis.

Education and health indicators jointly encode a country's development stage so consistently that 12 variables alone reconstruct well-known global inequality tiers, suggesting these dimensions are deeply correlated and may require coordinated policy interventions.